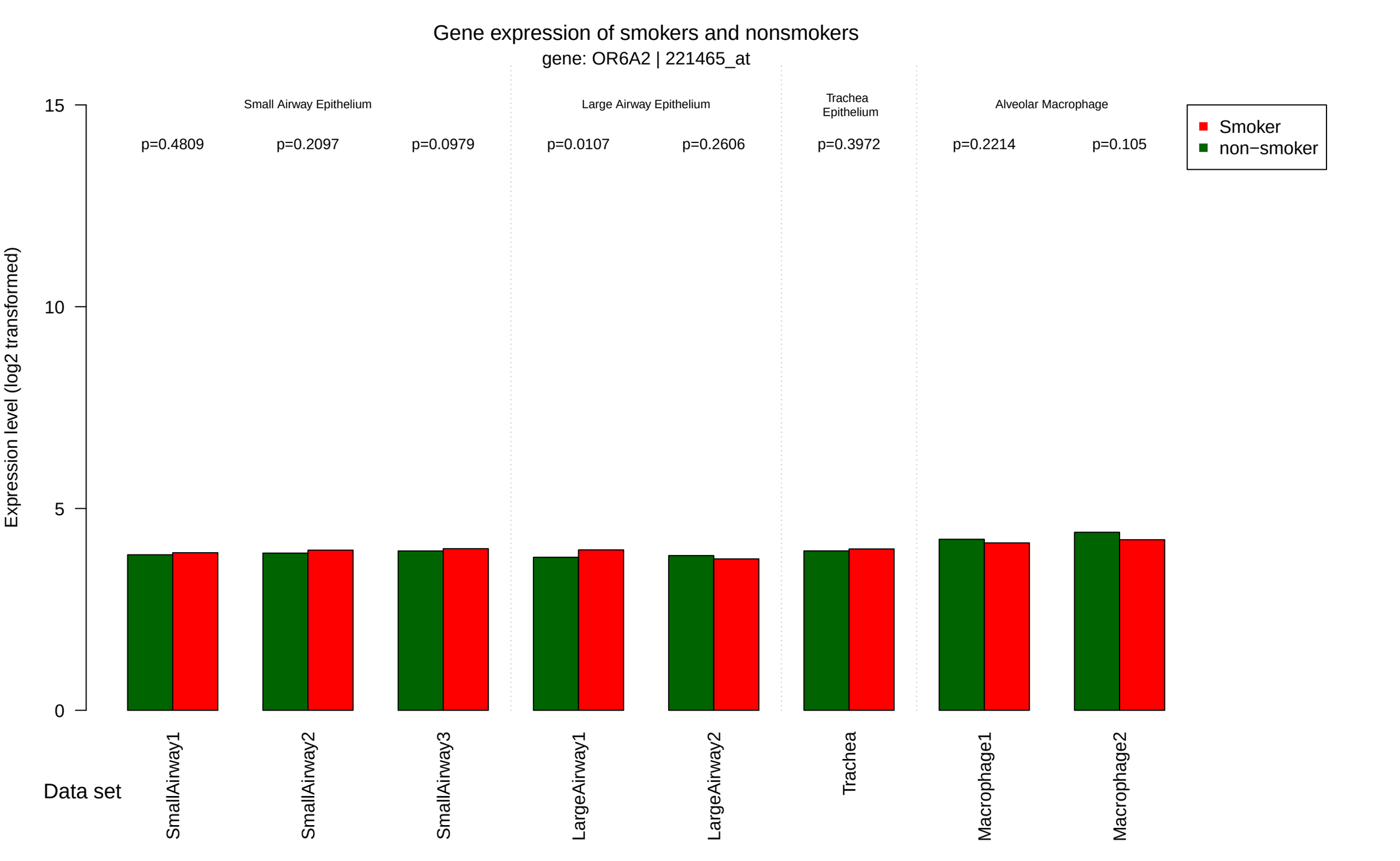
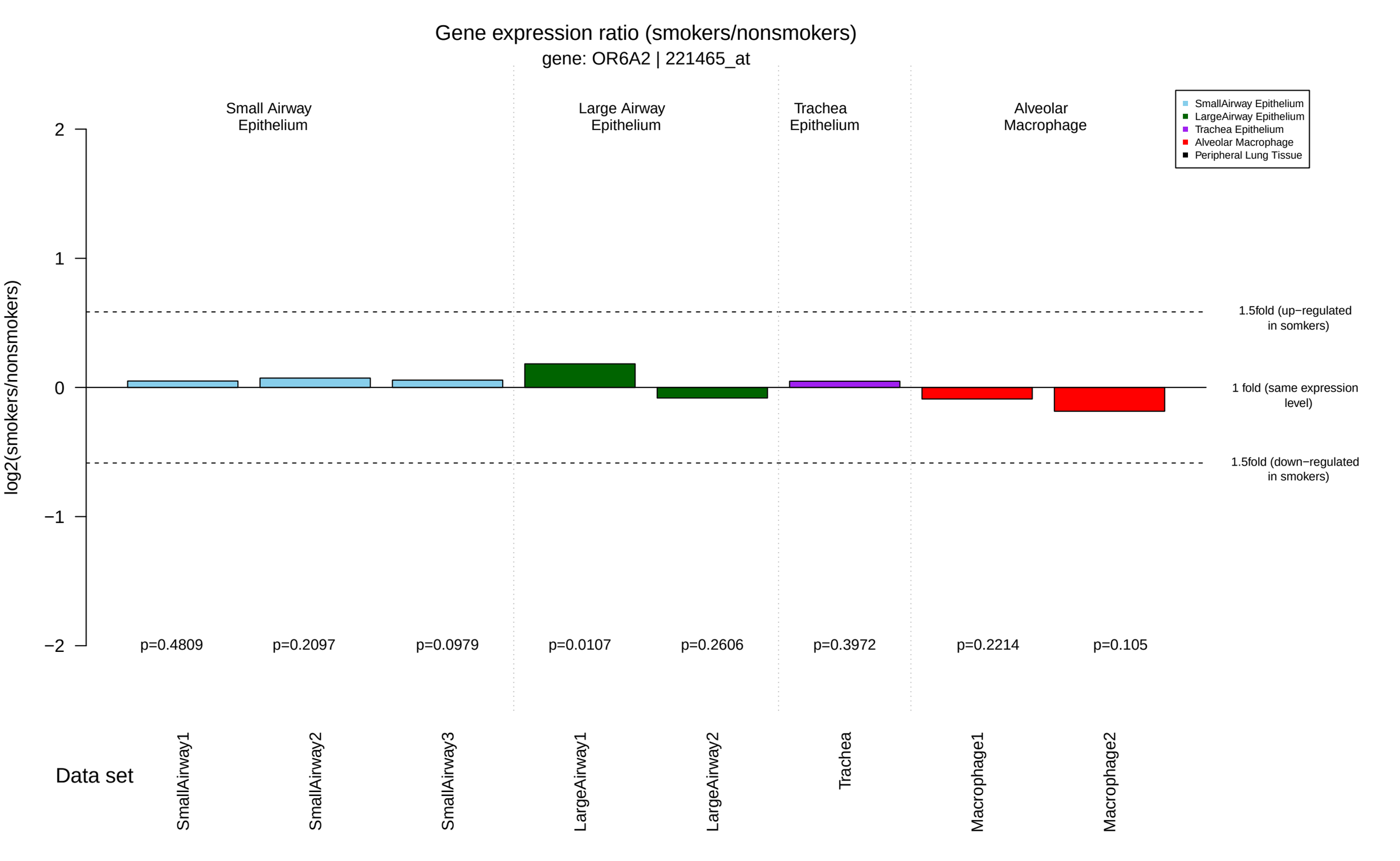
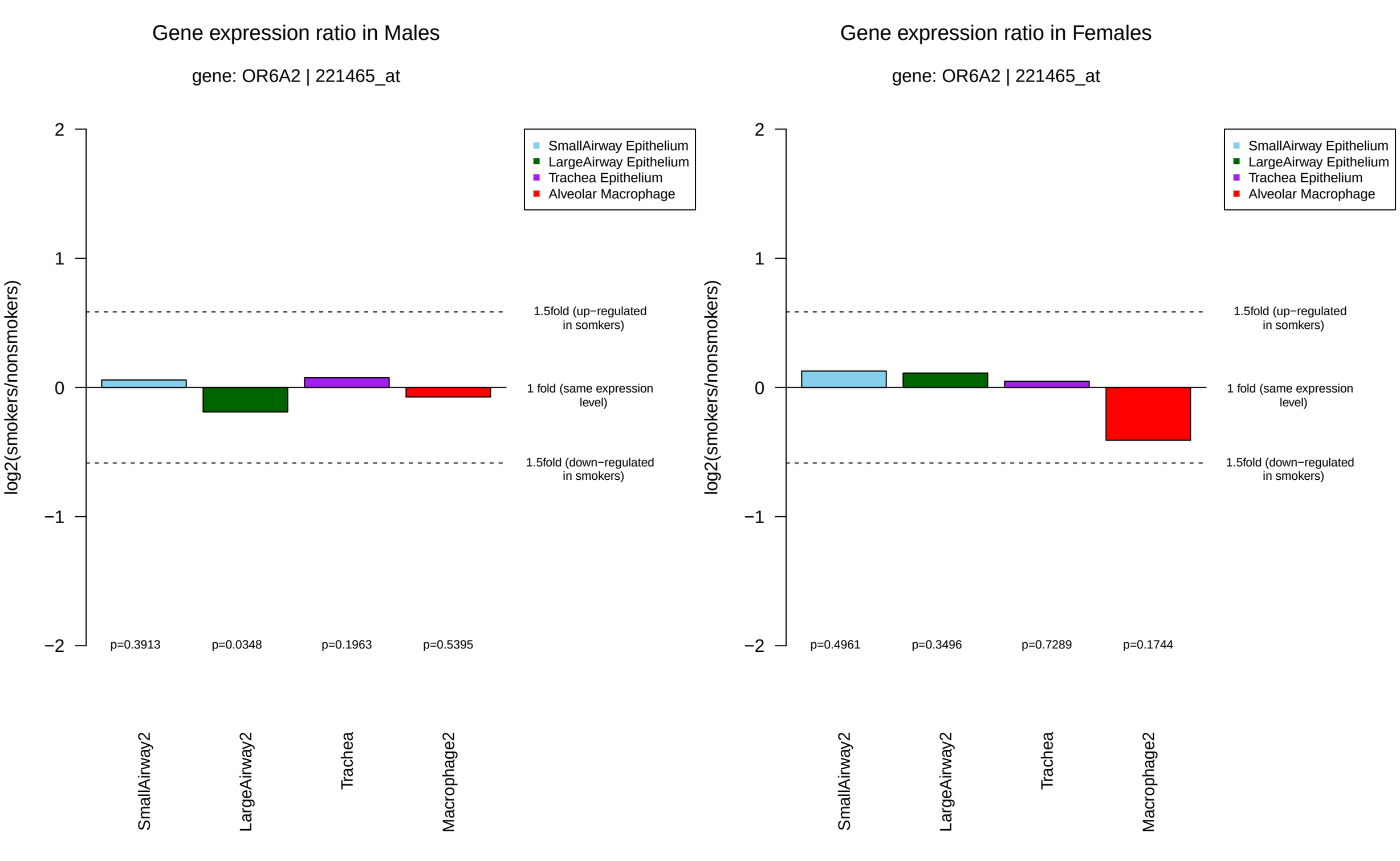
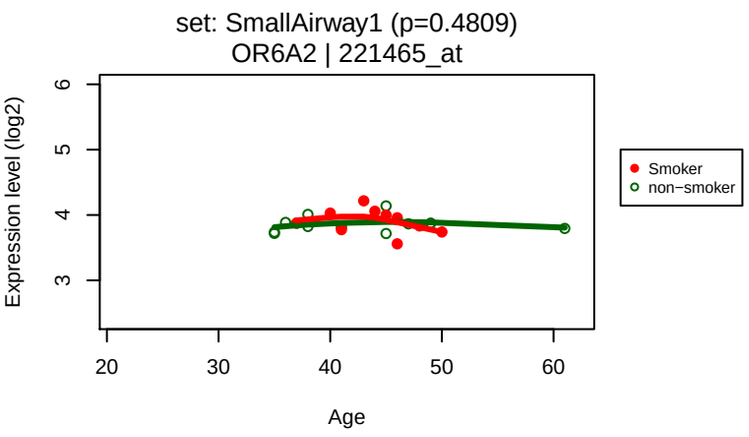
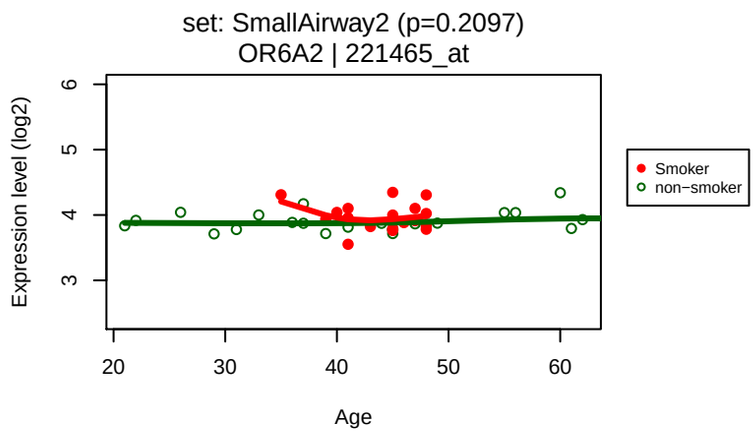
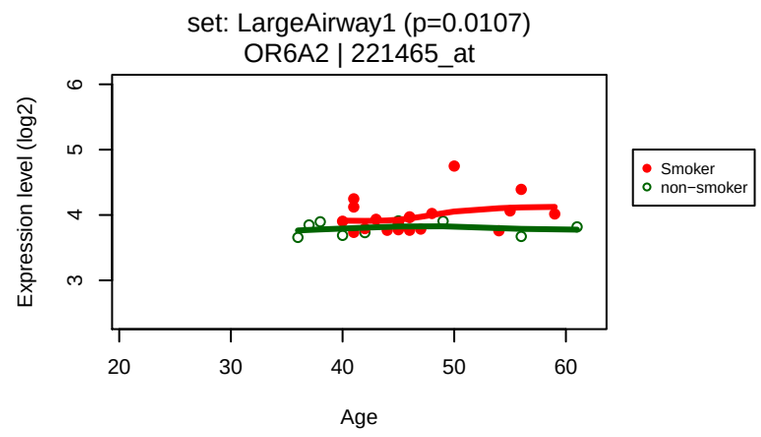

Gene expression of smokers and nonsmokers
gene: OR6A2 | 221465_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4809
p=0.2097
p=0.0979
p=0.0107
p=0.2606
p=0.3972
p=0.2214
p=0.105
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

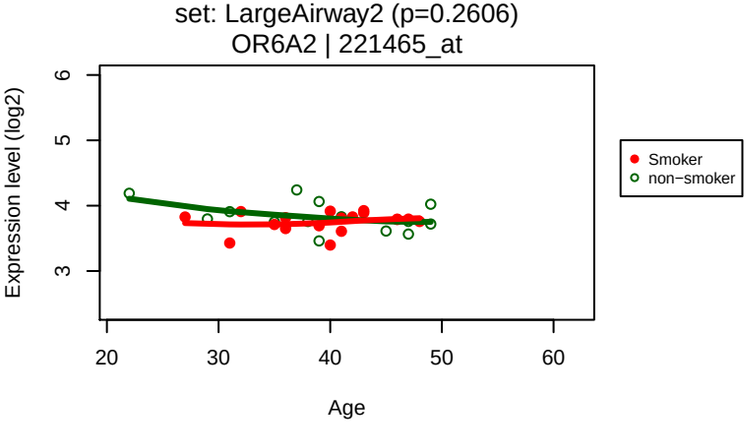

Gene expression ratio (smokers/nonsmokers)
gene: OR6A2 | 221465_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4809
p=0.2097
p=0.0979
p=0.0107
p=0.2606
p=0.3972
p=0.2214
p=0.105
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

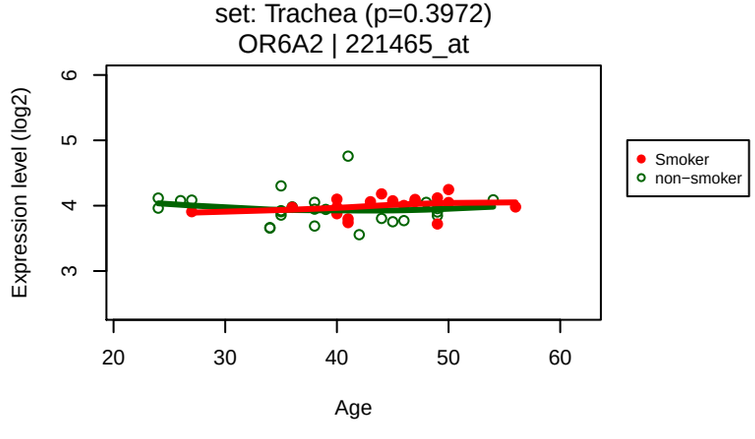

Gene expression ratio in Males
gene: OR6A2 | 221465_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3913
p=0.0348
p=0.1963
p=0.5395
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR6A2 | 221465_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4961
p=0.3496
p=0.7289
p=0.1744
SmallAirway2
LargeAirway2
Trachea
Macrophage2

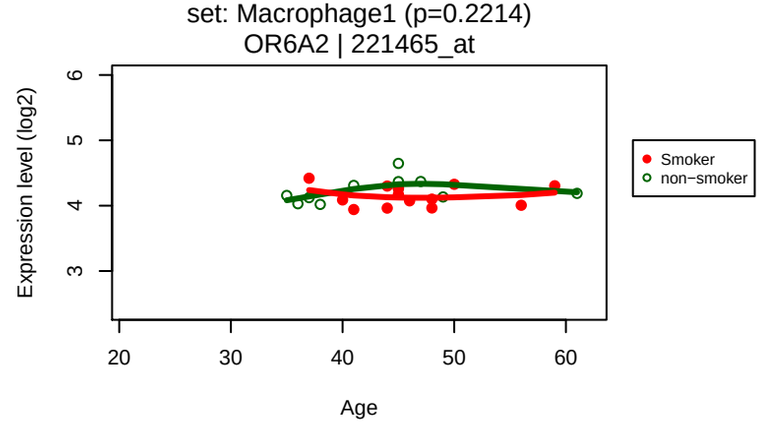

set: SmallAirway1 (p=0.4809)
OR6A2 | 221465_at
Expression level (log2)
Age
Smoker
non−smoker

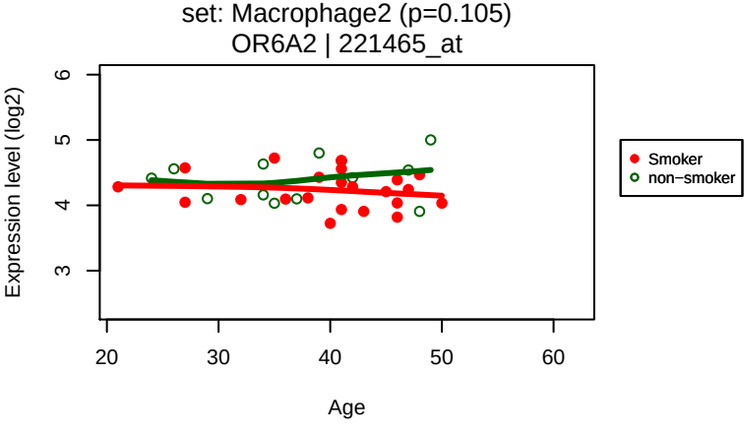

set: SmallAirway2 (p=0.2097)
OR6A2 | 221465_at
Expression level (log2)
Age
Smoker
non−smoker

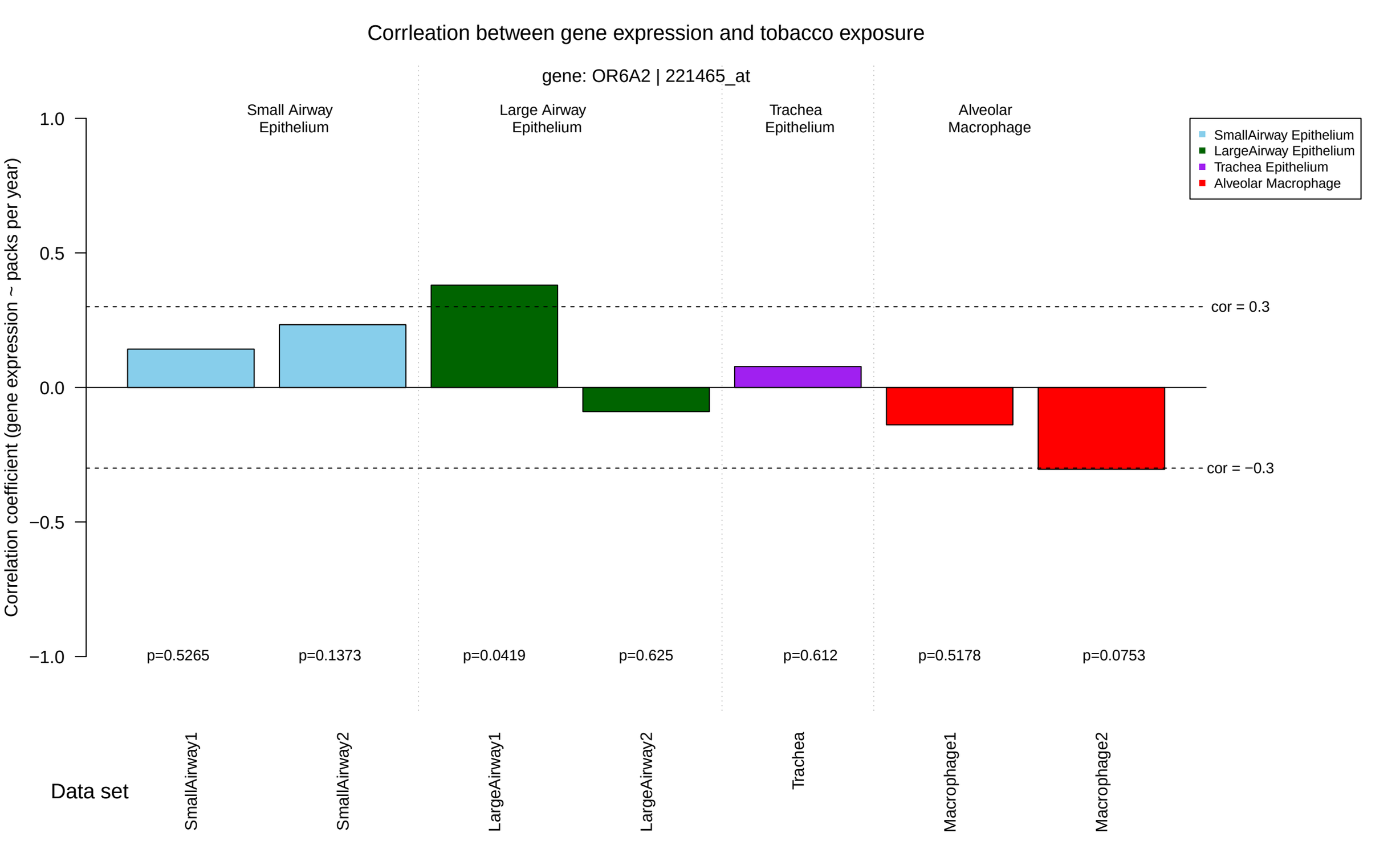

set: LargeAirway1 (p=0.0107)
OR6A2 | 221465_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2606)
OR6A2 | 221465_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3972)
OR6A2 | 221465_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2214)
OR6A2 | 221465_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.105)
OR6A2 | 221465_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR6A2 | 221465_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5265
p=0.1373
p=0.0419
p=0.625
p=0.612
p=0.5178
p=0.0753
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2